# Session 01 - Your First Encrypted Message - Tests

Python Contoso Vault - 1 November 2026

## Part 1 - Core: shift.py

### Goal

`shift.py` reads a message and a key, encrypts the message, then decrypts it again to prove the round trip works. Letters keep their case, and every other character stays the same.

Run `python shift.py` in the terminal. Type the message and press Enter, then type the key and press Enter.

### Instructions

1. Read a message as a string.
2. Read a key from 0 to 25 as an integer.
3. Shift each letter of the message forward by the key, within its own case.
4. Wrap any shifted letter that passes the end of its alphabet.
5. Leave every other character as it is.
6. Print the ciphertext.
7. Shift each letter of the ciphertext back by the key.
8. Wrap any shifted letter that passes the start of its alphabet.
9. Print the plaintext.
10. Prove that the plaintext matches the message.

### Constraints

- Show no prompt text when reading input, so the output holds only the required lines.
- The output is exactly two lines, in this order, with this exact wording and one space after each colon.
- The code contains no `%`.
- The code contains an `assert`.

### Input validation

Not required. Assume the key is always a whole number from 0 to 25. Any message is valid, including an empty one.

### Test plan

| Test | Message | Key | Expected output | Actual output | Pass |
|---|---|---|---|---|---|
| 1 | `Meet at Contoso HQ!` | 3 | `Ciphertext: Phhw dw Frqwrvr KT!`<br>`Plaintext: Meet at Contoso HQ!` | | |
| 2 | `Zebra zone 42, room 7.` | 25 | `Ciphertext: Ydaqz ynmd 42, qnnl 7.`<br>`Plaintext: Zebra zone 42, room 7.` | | |
| 3 | `Contoso HQ` | 0 | `Ciphertext: Contoso HQ`<br>`Plaintext: Contoso HQ` | | |
| 4 | `Contoso HQ` | 1 | `Ciphertext: Dpouptp IR`<br>`Plaintext: Contoso HQ` | | |
| 5 | `Contoso HQ` | 13 | `Ciphertext: Pbagbfb UD`<br>`Plaintext: Contoso HQ` | | |
| 6 | `Contoso HQ` | 25 | `Ciphertext: Bnmsnrn GP`<br>`Plaintext: Contoso HQ` | | |
| 7 | `2026-11-01, 9:00!` | 7 | `Ciphertext: 2026-11-01, 9:00!`<br>`Plaintext: 2026-11-01, 9:00!` | | |
| 8 | `mIxEd CaSe` | 4 | `Ciphertext: qMbIh GeWi`<br>`Plaintext: mIxEd CaSe` | | |
| 9 | empty line | 5 | `Ciphertext:`<br>`Plaintext:` | | |

### Done criteria

- [ ] Every test plan row prints exactly the expected lines.
- [ ] A search of the file for `%` finds nothing.
- [ ] The script runs from the terminal with `python shift.py`.
- [ ] Variables carry type hints.

# Part 2 - Challenge: crack_caesar.py

## Goal

`crack_caesar.py` reads a ciphertext and finds the key that reveals the word `"contoso"`, in any case.

Run `python crack_caesar.py` in the terminal. Type the ciphertext and press Enter.

## Instructions

1. Read a ciphertext as a string.
2. Try each key from 0 to 25, as an integer, in turn.
3. Decrypt the ciphertext with the current key, as in `shift.py`.
4. Stop at the first plaintext that contains `"contoso"` in any case.
5. Print the key.
6. Print the plaintext.
7. Report that no key was found if no key works.

## Constraints

- Show no prompt text when reading input, so the output holds only the required lines.
- A found key prints exactly two lines, `Key: <n>` then `Plaintext: <text>`. A miss prints exactly one line, `No key found`.
- The printed key is the one that turned the plaintext into the ciphertext.
- The code contains no `%`.

## Input validation

Not required. Any ciphertext is valid, including an empty line.

## Test plan

| Test | Ciphertext | Expected output | Actual output | Pass |
|---|---|---|---|---|
| 1 | `FRQWRVR PHHWV DW QRRQ` | `Key: 3`<br>`Plaintext: CONTOSO MEETS AT NOON` | | |
| 2 | `Sbujo ha uvvu` | `No key found` | | |
| 3 | `CONTOSO PAYROLL` | `Key: 0`<br>`Plaintext: CONTOSO PAYROLL` | | |
| 4 | `BNMSNRN OZXQNKK` | `Key: 25`<br>`Plaintext: CONTOSO PAYROLL` | | |
| 5 | `iutzuyu vgexurr oy jak` | `Key: 6`<br>`Plaintext: contoso payroll is due` | | |
| 6 | `Lxgw max vhgMHLh unwzxm` | `Key: 19`<br>`Plaintext: Send the conTOSo budget` | | |
| 7 | empty line | `No key found` | | |

## Done criteria

- [ ] Every test plan row prints exactly the expected lines.
- [ ] The decrypt logic matches `shift.py` on the same key.
- [ ] Variables carry type hints.